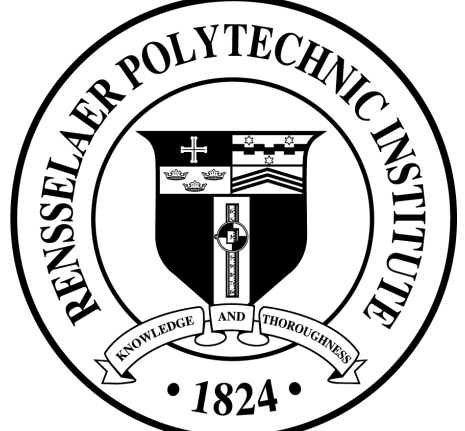




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I. Motivation and Background

Future space telescopes will rely on **real-time wavefront sensing and control (HOWFSC)** to achieve the high contrast ratios necessary to directly image Earth-like exoplanets around Sun-like stars. However, the unique radiation and resource-constrained environment of space prevents the use of high-performance computers typically employed in ground-based applications. **Current-generation radiation-hardened CPUs are unlikely to meet the computational demands of HOWFSC** on future missions like the *Habitable Worlds Observatory* (HWO)—**making FPGAs a compelling alternative** due to their deterministic performance and energy efficiency. One of HWO's key goals is to achieve a contrast ratio of 10^{-10} .

IV. Data-Flow Scheduling

We design acceleration kernels using a **dataflow model** leveraging pipelined execution, instruction-level parallelism, and optimized data scheduling and memory access.

For example, the **Angular Spectrum Method** kernel consists of three sequential blocks (F1, F2, F3) based on data layout and streaming direction (rows or columns), each block internally pipelined:

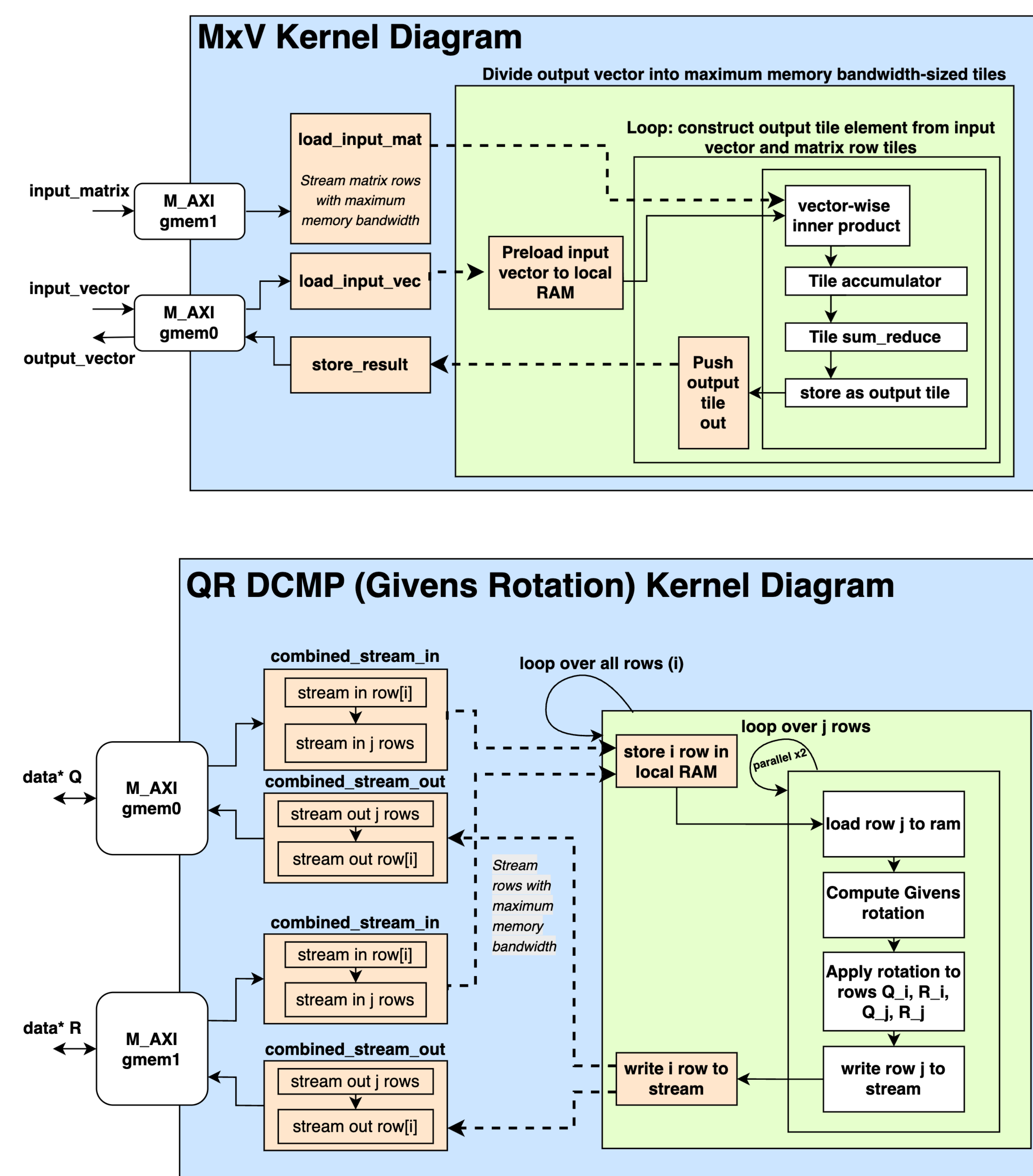
F1: Row-wise processing using **2 FFT IP cores** in pipelined streaming.

F2: Column-wise execution using **4 FFT IP cores** in parallel.

F3: Mirrors F1, completing the symmetry

- The second 2D FFT is reordered to match the column-major streaming, reducing transpositions through external memory.
- Data is loaded/stored with **maximum external memory bandwidth**, as already shifted to avoid redundant work.
- The **aperture function is constructed and applied on-the-fly** rather than loaded from memory, reducing I/O overhead and taking advantage of fast compute.

V. Kernels Flow Charts

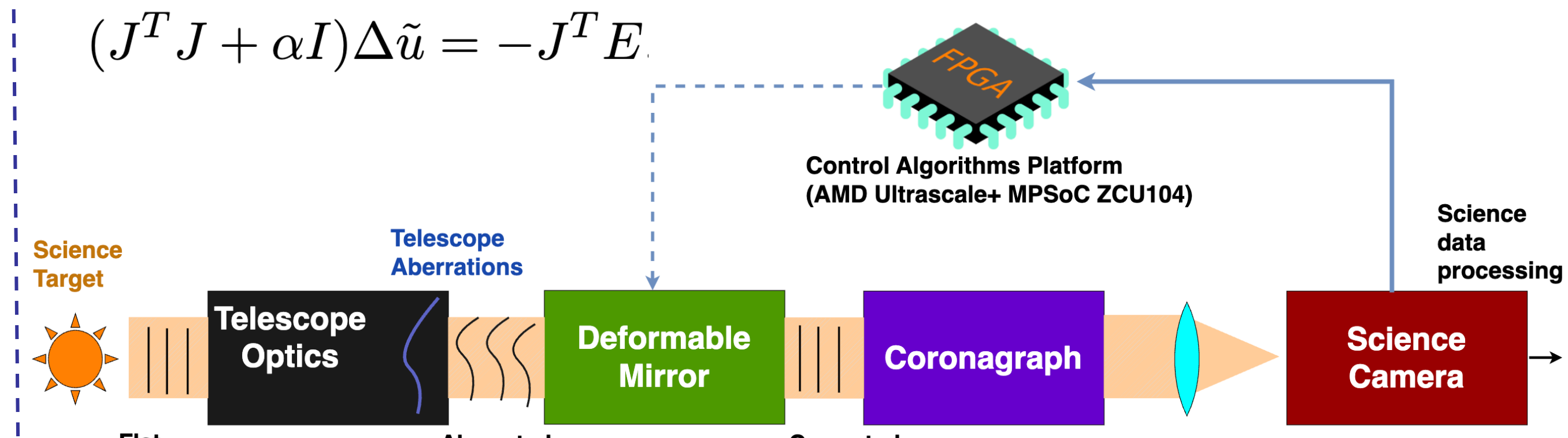


II. Control Algorithms

Electric Field Conjugation (EFC)

Computes the optimal actuator commands to minimize residual starlight intensity (electric field) in the dark zone of the image. The control update is computed by solving a regularized least-squares problem:

$$(J^T J + \alpha I) \Delta \tilde{u} = -J^T E$$



Jacobian Estimation

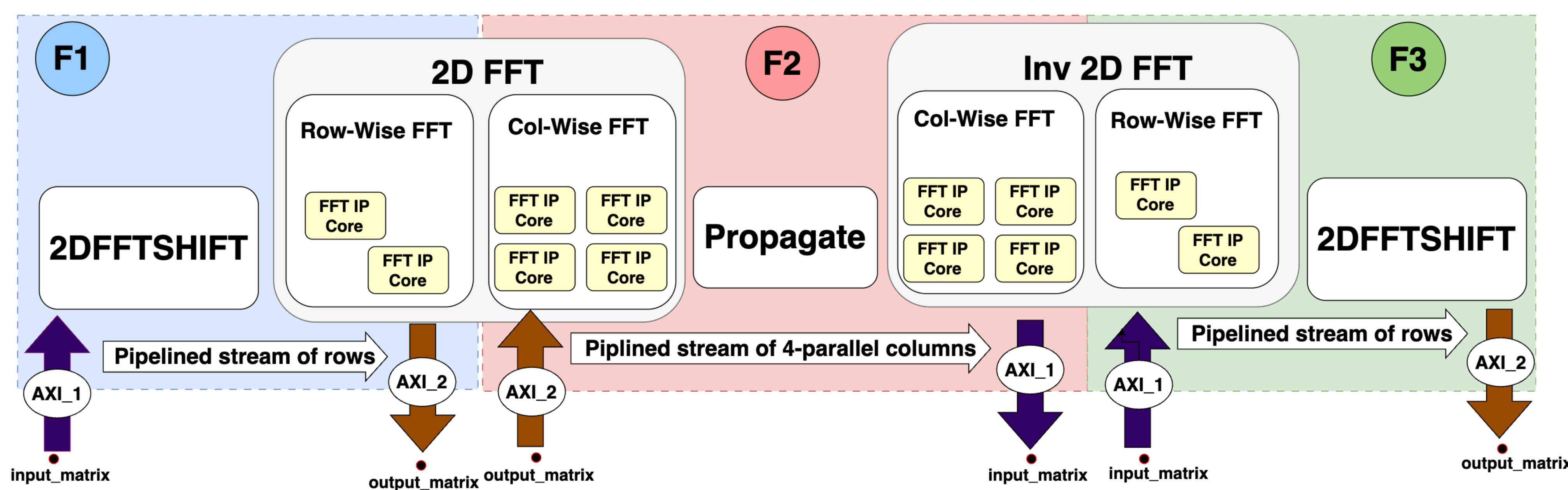
Determines the sensitivity of the image-plane electric field to each deformable mirror (DM) actuator. This is typically estimated using optical propagation models.

III. Our Contributions

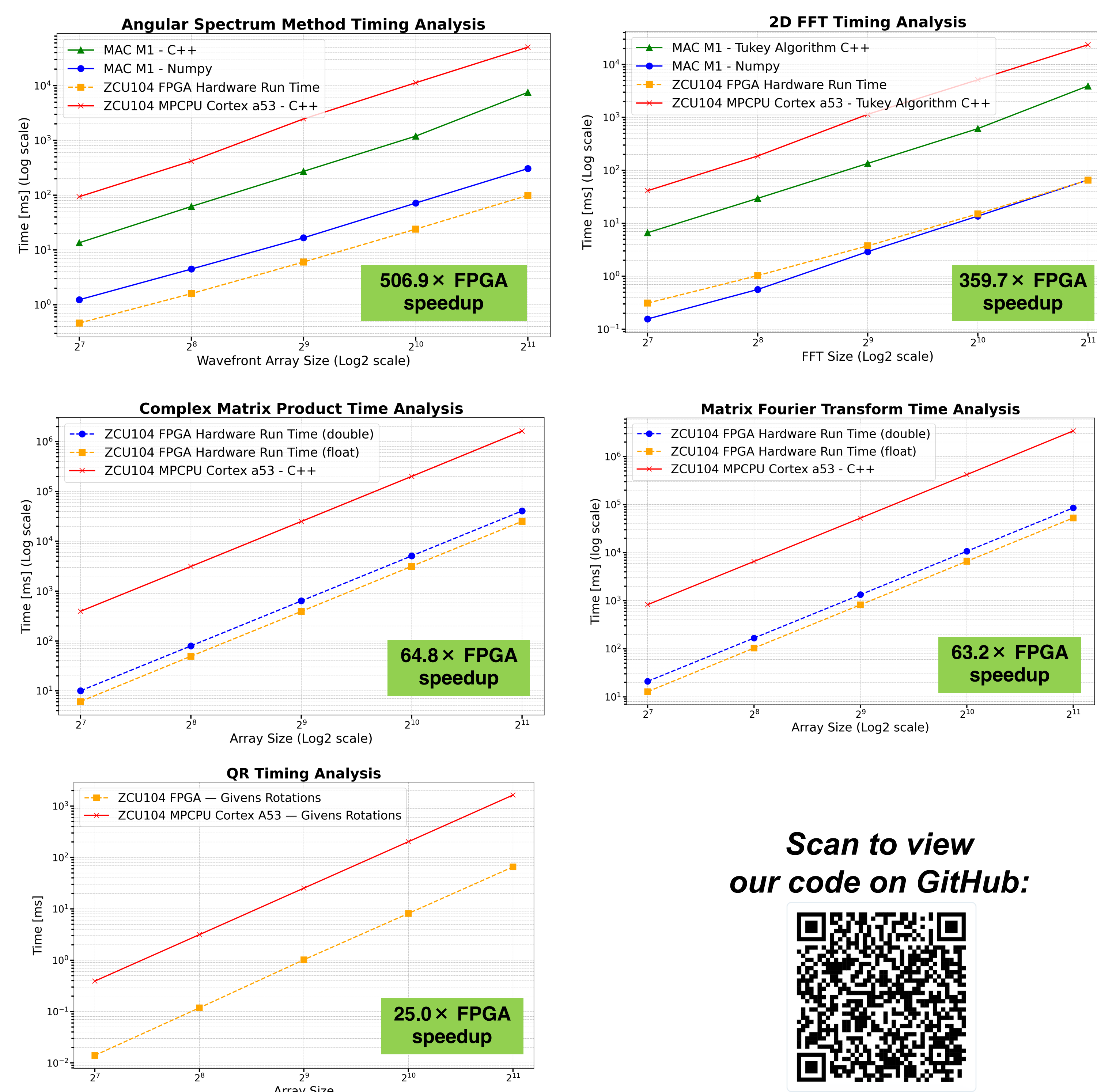
We identify and accelerate computational bottlenecks in HOWFSC using **custom-designed FPGA compute kernels**, targeting the **AMD Zynq UltraScale+ MPSoC ZCU104** platform. Our work:

- Designs **High Level Synthesis (HLS)** based hardware kernels for key operations.
- Integrates these kernels into a real-time demonstration system.
- Benchmarks the speedup over baseline **Arm Cortex-A53 CPU** implementations.
- Application development tools: AMD Vitis, Petalinux OS, Xilinx Runtime Library (XRT).

Operation Kernel	Algorithm Usage
2D FFT	Jacobian Estimation, Angular Spectrum Method
Angular Spectrum Method	Jacobian Estimation
Matrix-Matrix Multiplication	Jacobian Estimation, EFC, MFT
MFT	Jacobian Estimation
Matrix-Vector Multiplication	Jacobian Estimation
QR Decomposition	EFC



VI. Timing Plots



Speedups: FPGA vs. ZCU104 CPU (Arm Cortex-A53).
Working set of size $n = 2048$ of floating-points.

VII. Conclusion

FPGAs can significantly outperform CPUs

- High cost of increased development complexity.
- Dense linear algebra routines are limited by external memory bandwidth not compute resources
- FPGA performance increased with working set as processor caches become less effective.

Operation Kernel	FPGA Speedup
Angular Spectrum Method	506.9x
2D FFT	359.7x
Matrix-Matrix Multiplication	64.8x
Matrix Fourier Transform (MFT)	63.2x
QR Decomposition	25.0x

VIII. References

- Belsten, N., Embedded Computing for Wavefront Control on Future Space Telescopes, ph.d. thesis, Massachusetts Institute of Technology, Cambridge, MA (June 2025). Licensed under CC BY-SA 4.0.
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Scan to view
our code on **GitHub**:

